

# *“Is Stephen Curry really a guard? A new perspective on player typologies using functional data analysis”*: Reviewer Responses

## **GENERAL/ OVERALL RESPONSES**

We thank the associate editor and the two reviewers for their thorough and detailed review of our manuscript. We have addressed each review point by point, making appropriate changes in the revised manuscript. As a result, we believe that the revised manuscript is greatly improved. Responses to each of the review points are contained in the remainder of this document.

## ASSOCIATE EDITOR

### COMMENT # 0.1

*The author has effectively addressed some of the comments from the initial submission. Two new referees have reviewed the paper and give several suggestions for further revision. In addition, the authors should consider the following points.*

#### Reply:

We thank the associate editor for the careful consideration of our manuscript. We have addressed their points and those of the two reviewers and greatly improved the manuscript as a result. Our responses are presented in a point-by-point fashion below.

### COMMENT # 0.2

*The MFPCA approach was described more clearly in the revision, which leads to an important point about visualization. In equation (2), the eigenfunctions  $\phi_k(s, t) \in \mathbb{R}^2$  are bivariate, with one component corresponding to made and the other to missed shots. Doing MFPCA instead of marginal FPCA is nice since variability in made and missed shot locations are taken together, yielding a single set of scores for each player to cluster. However, there is no indication in the figures regarding how much each eigenfunction loads onto made and missed shots. In other words, it is possible that  $\phi_1(s, t)$  loads mostly onto made shots and almost none onto missed shots, or vice versa, but there is no way to tell this from the current analysis because the normalization (a linear transformation, I assume) is applied separately to every panel in each figure. It would be more informative to apply this transformation to the combined data across made and missed shots - that is, normalization is applied to each column. Also, it would be helpful to report how much each component contributes to the variability explained in made/missed shots separately, in addition to the joint figures currently reported.*

#### Reply:

### COMMENT # 0.3

*Although the author has argued that lack of consensus in a suitable spatial grid leads to subjectivity in previous analyses, there is no apparent justification for the choice of a  $201 \times 201$ , equally spaced grid for the MFPCA. In addition, the density estimation induces further subjectivity by choice of a bandwidth. Is there a way to assess sensitivity of the results and interpretations to these choices? It seems the choice is no less arbitrary for density representations*

compared to previous methods.

**Reply:**

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COMMENT # 0.4

*Related to the previous point, it is clear that densities are able to examine patterns without the influence of volume. However, how would removing volume information prove valuable in making performance-related decisions in terms of coaching, scouting, or training, for example? Are there other solutions that could be more effective, such as stratifying by volume rather than removing it altogether? Please address this more completely in the discussion.*

**Reply:**

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COMMENT # 0.5

*The choice of  $k = 5$  is clearly not data-driven. What would a true data-driven clustering look like? Certainly there are more than 5 shot pattern styles, so what information can we really gather from the clustering beyond observing that they clusters with  $k = 5$  are poorly aligned with player position? Also, although several metrics on the clustering are reported (ARI, Silhouette score), the relevance of the values is not discussed or their bearing on the reliability and impact of the clustering results.*

**Reply:**

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## REVIEWER # 1

### COMMENT # 1.1

#### *Abstract*

- *Overclaims novelty; FDA-based spatial modeling exists and differentiation from prior intensity-surface approaches is insufficiently clarified.*
- *No explicit statement of key methodological limitations (e.g., normalization removing volume/efficiency information).*
- *Practical contribution vs methodological novelty not clearly separated.*

#### **Reply:**

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### COMMENT # 1.2

#### *Introduction and Background*

- *Motivation for density representation vs intensity-based models is weakly justified; discretization critique is not convincing since FDA also requires discretization.*
- *Limited positioning within modern representation learning or embedding-based player typologies.*
- *Missing discussion of role-dependent contextual factors (pace, lineup context, possessions, play types).*
- *Additional reference that could be added to the manuscript is: <https://www.mdpi.com/2504-4990/6/3/102> and <https://www.mdpi.com/2078-2489/16/8/699>*

#### **Reply:**

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### COMMENT # 1.3

#### *Methodology*

- *Kernel density estimation choices (Gaussian kernel, Silverman bandwidth) insufficiently validated or sensitivity-tested.*
- *Grid resolution arbitrary without robustness analysis.*

- *Normalization removes shot volume and efficiency, limiting interpretability of components.*
- *MFPCA assumptions may violate density constraints (non-negativity, unit integral) — acknowledged but not resolved.*
- *Choice of  $k = 5$  clusters driven by court positions rather than data-driven criteria.*
- *Limited justification for k-medoids vs alternative clustering or manifold methods.*

**Reply:**

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#### COMMENT # 1.4

##### *Results*

- *Interpretation largely descriptive; lacks statistical validation or hypothesis testing.*
- *Low cluster agreement ( $ARI = 0.04\text{--}0.05$ ) not deeply analyzed — could indicate model inadequacy rather than novel insight.*
- *Silhouette scores suggest weak cluster structure but implications are under-discussed.*
- *No external validation (e.g., predictive performance, downstream task).*

**Reply:**

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#### COMMENT # 1.5

##### *Discussion*

- *Claims of interpretability and practical relevance not empirically demonstrated (e.g., coaching decisions, scouting outcomes).*
- *Limited comparison against baseline models or alternative representations.*
- *Absence of multimodal or contextual variables despite acknowledging importance.*

**Reply:**

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#### COMMENT # 1.6

##### *Conclusion and future work*

- *Future work suggestions are broad but not technically grounded (e.g., multimodal integration lacks methodological direction).*
- *Does not address scalability, computational complexity, or real-time applicability.*
- *Key methodological limitation (density normalization) insufficiently emphasized.*

#### **Reply:**

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#### COMMENT # 1.7

##### *Technical Writing*

- *Some redundancy in explaining FPCA concepts.*
- *Overly descriptive narrative in results; could benefit from more concise statistical framing.*
- *Terminology occasionally inconsistent between “shooting frequency,” “density,” and “intensity.”*
- *Figures interpretation sometimes qualitative rather than quantitative.*
- *Keywords are not precise and should be in alphabetical order.*

#### **Reply:**

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## REVIEWER # 2

### COMMENT # 2.1

*Alternative clustering approaches I think the authors should include the discussion of alternative clustering methods in the main text (or in the appendix).*

**Reply:**

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### COMMENT # 2.2

*Since the data span multiple seasons (2018-19 through 2023-24), shot selection can change across seasons (and also within a season), due to numerous reasons (e.g., changes in team, coaching, usage rate, playing style, etc.). This is perhaps outside the scope of the paper, but I'm interested in seeing a discussion on whether your method can be extended to handle this. I think having a couple of paragraphs on this in the Discussion would make the paper more intriguing to interested readers. For example, young players may enter the league without a reliable 3-point shot and therefore prefer scoring near the basket. Over time, they may develop a consistent 3-point shot and become more comfortable taking 3s. A similar pattern may apply to centers. Players such as Al Horford and Brook Lopez once favored post-ups and shots around the rim but later expanded their range and became good at 3-point shooting. (These are just examples of the top of my head; these developments by Horford and Lopez occurred before the sample of seasons considered in this paper.) Alternatively, have you also considered clustering at the player-season level instead of just one single aggregated shot chart for each player? Extending this to a player-season level could further validate your finding of empirical clusters having low agreement with official positions. For instance, it could help quantify the role/positional evolution of a player (related to what I described above).*

**Reply:**

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